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Shartaz Sajid Nahid

🐙 Github

🌐 shartazsajidnahid.github.io

Uttar Badda, Dhaka, Bangladesh

🔍 Google Scholar

RESEARCH INTERESTS

Machine Learning, Pattern Recognition, Low-dimensional Representation, Transfer Learning

EDUCATION

M.Sc. in Software Engineering, University of Dhaka
CGPA: 4.00/4.00

Nov 2025

B.Sc. in Software Engineering, University of Dhaka
CGPA: 3.96/4.00

Feb 2024

RESEARCH PUBLICATIONS

Shartaz Sajid Nahid, Md. Shariful Islam, Md. Arman Hossain, Muhammad Mahbub Alam, Mohammad Shoyaib, “OCMTL: Transfer learning by orthogonal core-extraction of a matrix and its application in cross-project defect prediction”, Alexandria Engineering Journal (**Elsevier, Q1, IF: 6.4**), Volume 133, 2025, Pages 498-516, ISSN 1110-0168. [Link to publication]

Md. Siam, **Shartaz Sajid Nahid**, Md. Arif Hasan, Md. Nurul Ahad Tawhid, and Kazi Muheymin-Us-Sakib, “DTM: Deterministic Approaches for Black-box Test Suite Minimization with Tree-based Similarity”, 32nd Asia-Pacific Software Engineering Conference 2025. (Accepted for publication) [Link to manuscript]

Md. Sabbir Hosen, **Shartaz Sajid Nahid**, Muhammad Asif Hossain Khan, and Ahmedul Kabir, “A Color-guided Trajectory-based Approach for Multi-worker Sewing Activity Recognition”, Journal of Engineering, 2026. (Under review) [Link to manuscript]

RESEARCH EXPERIENCE

Orthogonal Core Matrix for Transfer-learning | Graduate Researcher

2024 – 2025

Proposed two parameter-free, low-cost methods: (i) a subspace-learning method for variance-preserving latent representation extraction and (ii) a transfer-learning method for learning shared orthogonal cores between source and target domains, using iterative SVD updates with convergence guarantee. (Masters Thesis, published in Alexandria Engineering Journal (Elsevier, Q1, IF: 6.4))

A Low-cost Vision-based System to Improve Efficiency of Assembly Lines in Ready made Garments Factories | Research Assistant

2024 – Present

Proposed a CCTV-based framework for sewing activity recognition that detects the hand via a worn color-band marker, constructs motion trajectories, extracts trajectory features, and classifies activity (manuscript under review). Currently working toward a tensor-based anomaly-detection formulation to identify the sewing activities. Funded by the University Grants Commission, Government of Bangladesh.

Using AI and LLMs to automate data extraction and analyses from unstructured government documents | Research Assistant

2024 – 2025

Combined LLM fine-tuning, prompt-based extraction, and traditional NLP methods to extract structured information from unstructured government reports collected across multiple regions. Funded by Government of Bangladesh (confidential under N.D.A.).

Automated Information Extraction and Visualization of Road Accident News | Undergraduate Researcher

2023

Developed a web application that automatically scrapes Bangla news articles on road accidents from online newspapers, extracts key information, classifies them and visualizes for pattern analysis. (Honors Thesis)

Tactile Vision - Braille to Bangla | Undergraduate Researcher2023

Developed an application that takes scanned Bangla Braille images as input, identifies Braille text from the images, and converts Braille text to Bangla text.

TEACHING EXPERIENCE

East West University Lecturer in CSE [Link to profile]	Dec 2025 – Present
Courses: Statistics for Data Science, Machine Learning, Object-Oriented Concepts, Software Engineering	
University of Liberal Arts Bangladesh Lecturer (Adjunct) in CSE	Oct 2025 – Dec 2025
Courses: Artificial Intelligence & Machine Learning, Big Data Analytics	
Institute of Information Technology, University of Dhaka Instructor in Short Course Training Program	May 2024 – Sep 2025
Courses: Python programming, Office Applications for Smart Office	

PROFESSIONAL EXPERIENCE

IEEE 3rd International Conference on Computing, Applications and Systems Reviewer	2026
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GRANTS & AWARDS

Research Fellowship for Master’s Thesis ICT Division, Ministry of Posts, Telecommunications and Information Technology, Bangladesh	2024 – 2025
Talentpool Scholarship for Honors Government of Bangladesh	2024 – 2025
Champion at Yearly Competitive Programming Contest IIT, University of Dhaka	2019

TECHNICAL SKILLS

Languages: MATLAB, Python, C, C++, Java, JavaScript, PHP, Latex
ML Frameworks & Data Analysis: MATLAB Statistics and ML Toolbox, Scikit-Learn, TensorFlow, HuggingFace, Pandas, NumPy, Matplotlib, Seaborn
Web Technologies: Node.js, Express.js, FastAPI, Laravel, Angular, React
Database Technologies: MySQL, MongoDB

EXTRACURRICULAR ACTIVITIES

Player & Captain (2024-2025), IITDU Football Team	2019 – 2025
Player, IITDU Cricket Team	2019

REFERENCE

Dr. Mohammad Shoyaib Professor Institute of Information Technology University of Dhaka shoyaib@iit.du.ac.bd	Dr. Ahmedul Kabir Associate Professor Institute of Information Technology University of Dhaka kabir@iit.du.ac.bd
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